

Higashiyama & Kamo River, Kyoto

Then-and-now aerial photographs, how the ground was made, and the disasters people chose to remember — compiled automatically from public Japanese government data.



東山・鴨川（京都市東山区）

Study area	2.2 km × 1.4 km around 34.9990° N, 135.7790° E
Oldest aerial photograph	1945–1950
Aerial series in this brief	1945–1950, 1961–1969, 1974–1978, 1979–1983, 2020
Main natural landform	Alluvial fan (42%)
Former river channels	none mapped
Filled, reclaimed or drained land	none mapped
Former sea, river or pond	none mapped
Meiji-era paddies, marsh and flats	sand and gravel 2.0%
Disaster memorials within 3 km	1 (other 1)

What this brief is. A reading aid for comparing the same ground across decades. It shows what the photographs and GSI's land surveys record; it does not describe current conditions, access or opening hours.

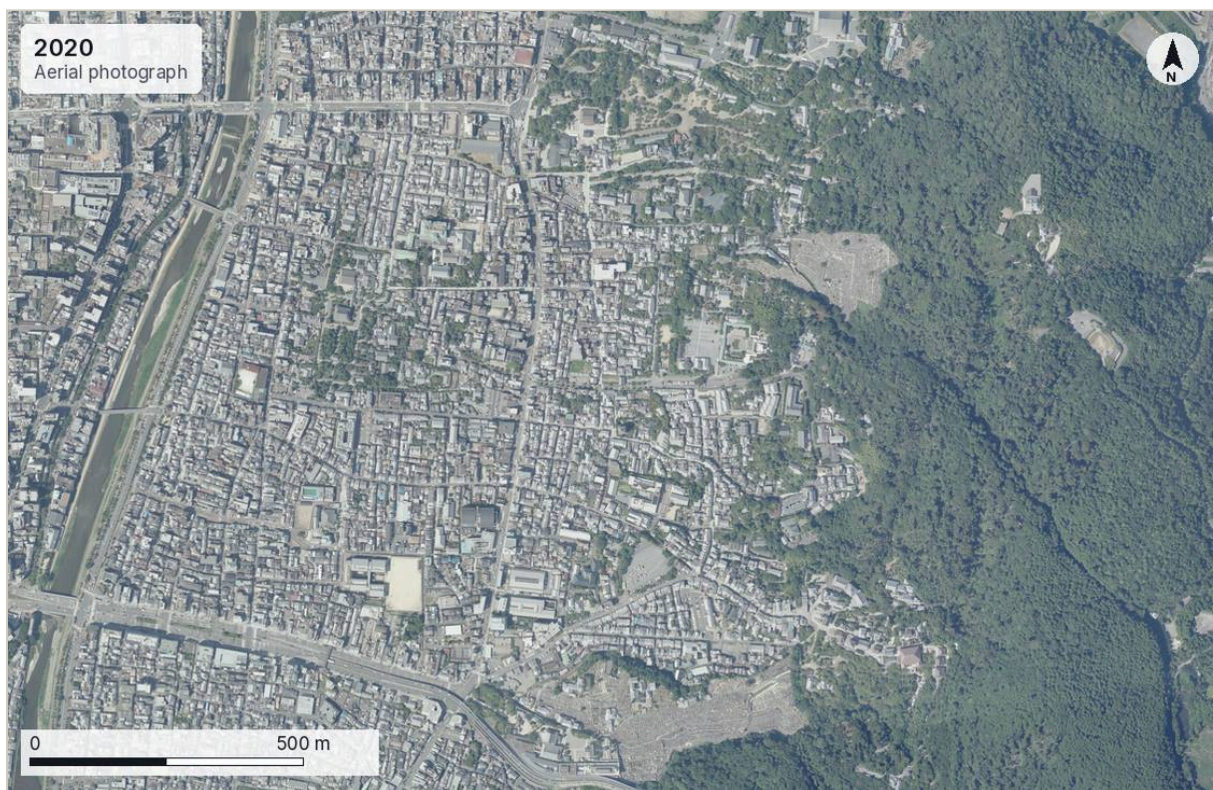
What it is not. Not a hazard assessment or a property survey. Landform classes describe typical tendencies of the ground; for present-day risk, read the official hazard map of the municipality.

Edition 2026-09-27. Aerial photographs, landform and Meiji-era lowland data: Geospatial Information Authority of Japan (GSI), processed by Japan Time Atlas. Memorial data: GSI Natural Disaster Memorials, summarised and translated by Japan Time Atlas.

The same ground, 1945–1950 and 2020



1945–1950 aerial photograph (US military photography) — GSI Tiles *ort_USA10*, cropped and annotated by Japan Time Atlas. North is up; the frame is identical in both images.



2020 annual orthophoto — GSI Tiles *nendophoto2020*, cropped and annotated by Japan Time Atlas. Capture dates within a mosaic can differ from place to place.

How to compare. Anchor on features that rarely move — river banks, rail lines, shrine and temple grounds, main roads — then look for what changed around them. Differences in tone come from film, camera and season, not from the land itself.

The decades in between

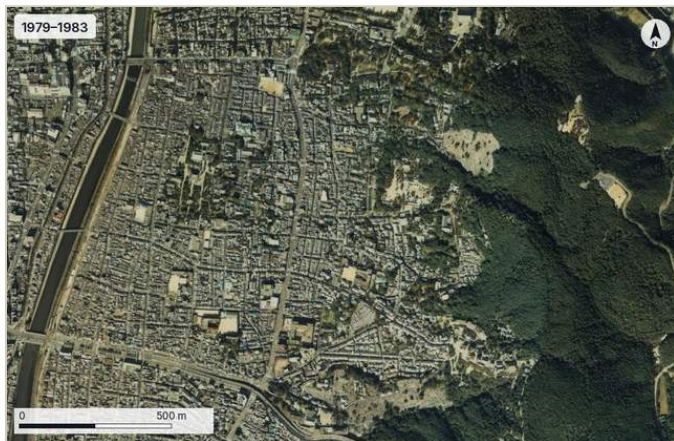
Every further GSI series that covers the same frame, oldest first. Each series is a compilation: photographs within one series were taken on different dates, which is why seams and changes of tone can appear inside a frame.



1961–1969 · GSI Tiles *ort_old10*, processed

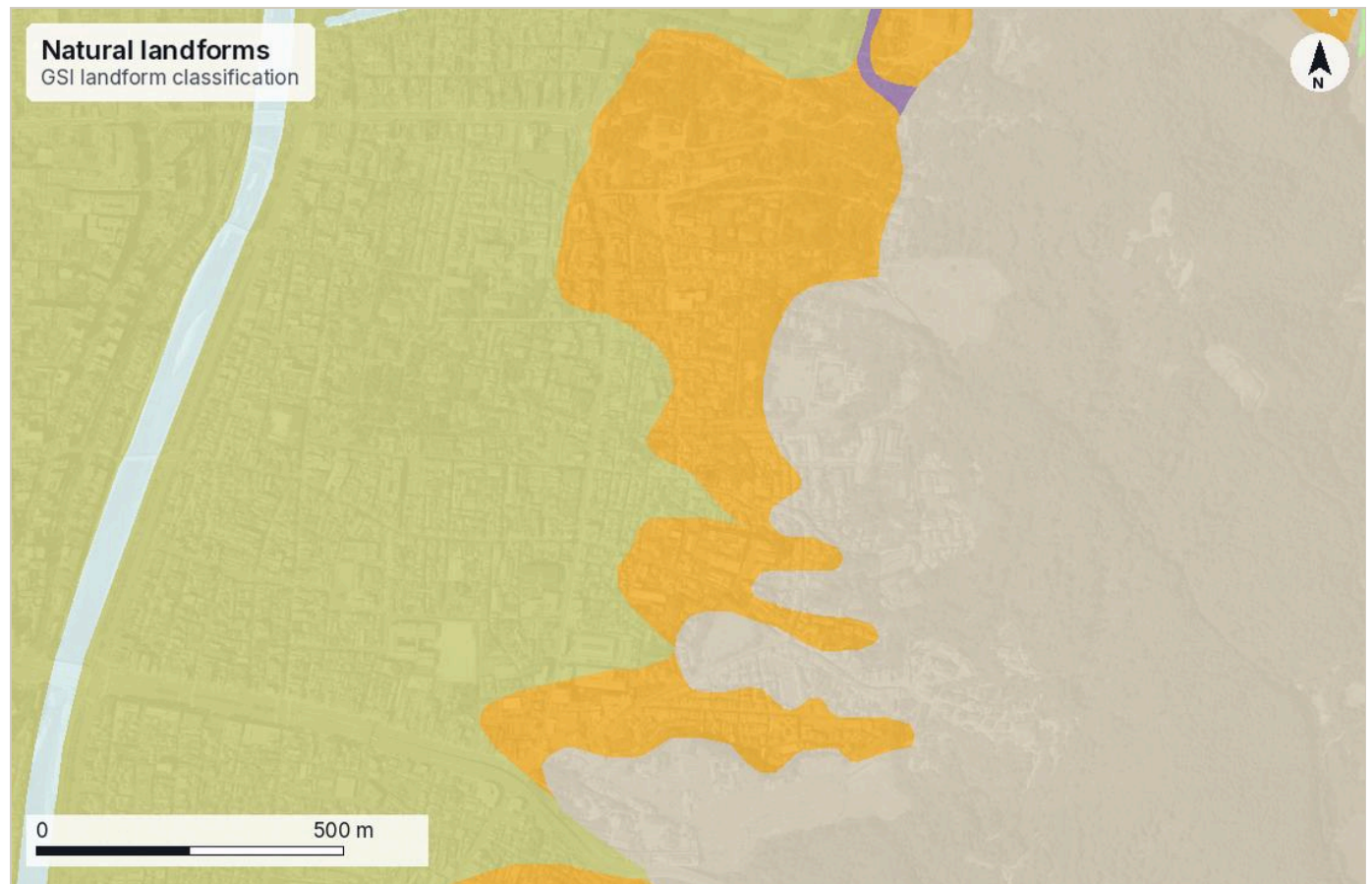


1974–1978 · GSI Tiles *gazo1*, processed



1979–1983 · GSI Tiles *gazo2*, processed

Rivers, shorelines and terraces



Natural landforms as classified by GSI, over a recent aerial photograph in gray. Dark blue outlines mark former river channels.



What the main landforms mean

Alluvial fan (42%). Gently sloping ground spreading fan-like from a valley mouth, built from sediment dropped by floods. Floodwater from the mountains, and debris flows near the valley mouth. Relatively firm ground; a slight liquefaction tendency on the lower fan.

Mountain and hill slopes (42%). Ridges, valleys and fairly steep ground, including terrace-scarp slopes and volcanic ground. Heavy rain or earthquakes can trigger slope failures, debris flows and landslides.

Plateaus and terraces (13%). Flat ground standing in steps above its surroundings, left behind as the land around it was eroded. River flooding is rarely a threat unless the terrace is only slightly above the river; watch the slopes at its edges. Firm ground: shaking is relatively weak and liquefaction is unlikely.

Descriptions translated and condensed from GSI's landform-classification legend. They describe tendencies of each landform type, not the condition of any individual plot.

Fill, reclamation and the fields beneath the town



Hatching: man-made ground as classified by GSI. Teal outlines: former sea, river or pond. Dark blue outlines: former river channels.



GSI “Meiji-era lowland” data, traced from topographic maps of the 1880s–1900s; GSI warns that positions can be off by a considerable distance, especially in the Kanto and Kinki regions. Only low, wet or open ground was traced: towns, dry fields and forest of the period are left blank.

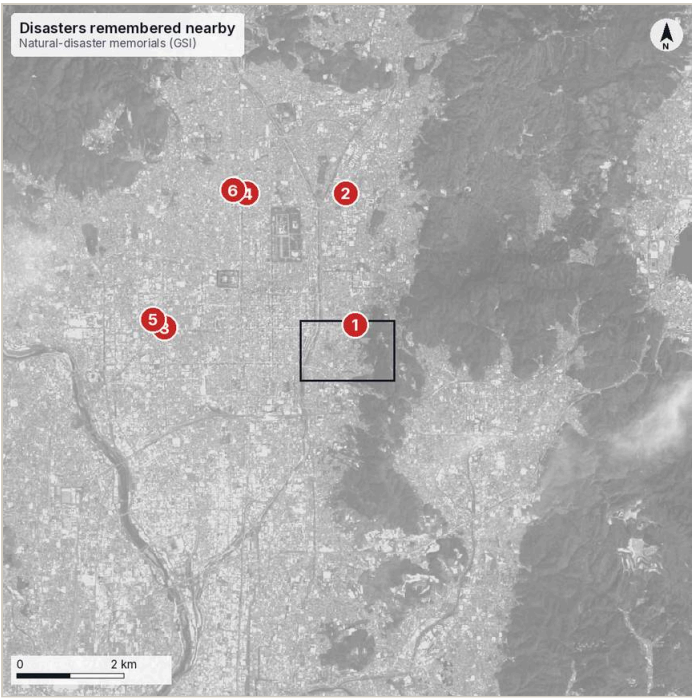
Man-made ground

No man-made ground mapped in this frame.

Meiji-era lowland

- Sand and gravel 2.0%
- River, pond or sea 1.6%

Memorials to natural disasters nearby



Numbered memorials; the black rectangle is the study area of this brief. Frame 16 km across.

Fewer than three memorials stand within 3 km of the study area, so the nearest ones within 8 km are listed.

Memorial	Disaster	Erected	Distance
1 Statue of the bond between teachers and pupils, for the 1934 Muroto Typhoon victims 師弟愛の像	Muroto Typhoon (21 Sep 1934) · other	1960	0.6 km
2 Stone record of the 1934 Muroto Typhoon damage at Yosei school 記念誌	Muroto Typhoon (21 Sep 1934) · other	1941	3.7 km
3 Memorial tower for the Junna school victims of the 1934 Muroto Typhoon 風災慰霊塔	Muroto Typhoon (21 Sep 1934) · other	1935	4.3 km
4 Memorial stone recording the 1935 flood inundation 昭和十年水害浸水被害記念碑	Kyoto flood (29 Jun 1935) · flood	unknown	4.3 km
5 Memorial to the 1934 Muroto Typhoon disaster at Junna school 風災記念碑	Muroto Typhoon (21 Sep 1934) · other	unknown	4.6 km
6 Memorial to the 41 pupils of Nishijin school killed by the 1934 Muroto Typhoon 風害記念碑	Muroto Typhoon (21 Sep 1934) · other	1935	4.6 km

What the memorials record

1. **Statue of the bond between teachers and pupils, for the 1934 Muroto Typhoon victims.** On 21 September 1934 the winds of the Muroto Typhoon, which raged across western Japan, brought down wooden buildings at several schools in and around Kyoto, killing 162 children and 4 teachers trapped beneath them. The present statue was rebuilt after the war.
2. **Stone record of the 1934 Muroto Typhoon damage at Yosei school.** When the Muroto Typhoon struck the Kinki region on 21 September 1934, Yosei Elementary School recorded 3 wooden classrooms destroyed, 3 half-destroyed, 4 tilted and 15 needing repair, and the roof of its indoor gymnasium badly damaged. None of the children was hurt.
3. **Memorial tower for the Junna school victims of the 1934 Muroto Typhoon.** On 21 September 1934 the winds of the Muroto Typhoon trapped more than 200 people under the collapsed building of Junna Elementary School (now Saiin Elementary School). Erected on the first anniversary, the tower bears the names of the 32 children and the teacher, Matsuura, who died.
4. **Memorial stone recording the 1935 flood inundation.** Before dawn on 29 June 1935, after days of heavy rain, all the rivers in Kyoto overflowed and more than 40,000 houses in the city were flooded. In the Ogawa district the Kamidachiuri-Ogawa stone bridge fell at 5 am, and houses along the stream collapsed, were swept away or were half-destroyed. The inscription records water four shaku (about 1.2 m) deep.

5. Memorial to the 1934 Muroto Typhoon disaster at Junna school. On 21 September 1934 the winds of the Muroto Typhoon, which raged across western Japan, trapped more than 200 people under the collapsed building of Junna Elementary School (now Saiin Elementary School); 32 children and a teacher, Matsuura, were killed.

6. Memorial to the 41 pupils of Nishijin school killed by the 1934 Muroto Typhoon. On 21 September 1934 the winds of the Muroto Typhoon, which raged across western Japan, brought down a building of Nishijin Elementary School, and 41 children trapped beneath it died.

Source: GSI Natural Disaster Memorials (自然災害伝承碑). English names and notes are Japan Time Atlas translations and summaries of GSI's Japanese descriptions; the memorials' own inscriptions may say more. "Erected" is the year GSI records for the monument.

Where every element comes from

- **Aerial photographs** (pp. 2–3) — Geospatial Information Authority of Japan (GSI), GSI Tiles: series *ort_USA10* (1945–1950, US military photography), *ort_old10* (1961–1969), *gazo1* (1974–1978), *gazo2* (1979–1983) and *nendophoto2020*. Cropped, resized and annotated by Japan Time Atlas. List of GSI Tiles: <https://maps.gsi.go.jp/development/ichiran.html>
- **Landform classification** (pp. 4–5) — GSI landform-classification vector tiles (natural and artificial landforms), compiled by GSI from its Land Condition Maps, Flood Control Terrain Classification Maps and vulnerable-landform surveys. Rendered, measured and translated into English by Japan Time Atlas. https://github.com/gsi-cyberjapan/experimental_landformclassification
- **Meiji-era lowland** (p. 5) — GSI “明治期の低湿地” tiles and legend (https://maps.gsi.go.jp/legend/lw_legend.pdf), measured and translated by Japan Time Atlas.
- **Natural disaster memorials** (p. 6) — GSI Natural Disaster Memorials (自然災害伝承碑), registered by municipalities, <https://www.gsi.go.jp/bousaichiri/denshouhi.html>; processed by Japan Time Atlas (selection by distance, English translation and summary). Memorial photographs are not reproduced.
- GSI content is used under the GSI Content Terms of Use, which follow the Public Data License v1.0 (PDL1.0, compatible with CC BY 4.0). Data retrieved 2026-09-27.
- **Outline of Japan** — Data of Japan / JapanPrefGeoJson (public domain), simplified.

Method

The study area is a fixed rectangle of 2.2 × 1.4 km. Every GSI photo series that covers at least 85% of it is included; the newest annual orthophoto covering at least 90% of it is used as “now”, with any gap filled from GSI's latest seamless mosaic. Landform shares are areas of the rectangle measured on GSI's polygons at zoom level 14, each class merged before measuring. Meiji-era shares count map pixels by legend color. Memorials are selected by straight-line distance from the center of the study area.

Limits

Historical series are mosaics of photographs taken on different dates; GSI publishes the flight date of each photograph. Landform data record conditions at the time of survey. Nothing in this brief shows present-day access, safety or opening hours.

Area Dossier. The same district with a 4–8 page English history, written from Japanese library sources and cited page by page.

Deep Research. A single hamlet or address, researched in libraries and archives.

Compiled by Japan Time Atlas; not produced, reviewed or endorsed by GSI, MLIT or any Japanese government agency. Not an official hazard map and not for statutory real-estate disclosure. Generated 2026-09-27 by an automated pipeline (research/lab-07/scripts/basic.py).